

Education

San Diego State University, Department of Computer Science

Bachelor of Science in Computer Science

Cumulative GPA: 3.83

Expected Graduation: May 2026

Relevant Coursework: Advanced Programming Languages • Operating Systems • Computer Architecture • Data Structures • Intro to Software Systems • Computer Organization • Discrete Mathematics • Linear Algebra

Proficient In: Python, C++, Object-Oriented Programming, GitLab CI/CD, Data Preprocessing, Git Version Control

Familiar With: Java, Go, HTML5, JavaScript, CSS, React, Shell Scripting, Data Analytics, Debugging and Testing, Computer Architecture, Docker, Software Development Lifecycle (SDLC), API Integration, Compiler Design

Work Experience

Glympse

Seattle, Washington

Location Sharing Technology

Software Engineering Intern

June 2024 – Present

- Developed Python and Golang scripts for file comparison, cutting memory usage by 25% via benchmarking.
- Migrated legacy code to a new repository, adding essential parameters and enhancing report functionality.
- Refactored and modernized a shell script report in Python, enhancing readability and maintainability.
- Built and implemented comprehensive unit tests to ensure the reliability and accuracy of the Python code.
- Deployed code using GitLab CI/CD pipelines, configuring environments with '.gitlab-ci.yml' and Docker.

San Diego State University

San Diego, California

Academic Institution

Student Research Assistant

February 2024 – June 2024

- Conducted extensive research under faculty in the Integrating Cyber Innovations with Physical World Lab.
- Investigated detailed medical device recalls to detect underlying problems related to software issues.
- Implemented efficient code to automate data extraction from the FDA recall website for relevant analysis.
- Engaged in collaborative problem-solving exercises to address research challenges and advance objectives.
- Utilized HTML5 to design a compelling and operational website that proficiently showcases research.

Treobytes

San Diego, California

STEM Education Program

Facilitator

March 2024 – May 2024

- Developed and customized innovative STEM lesson plans to suit the interests of diverse student groups.
- Facilitated intriguing educational sessions for students, delivering course content effectively and engagingly.
- Conducted meticulous and thorough quality assessments on STEM resources for accuracy and functionality.
- Demonstrated exceptional organizational and time management skills to balance multiple responsibilities.
- Assisted in the development of a coding course where students learned by creating their own video games.

Projects

StockScript

November 2024 – December 2024

- Designed and implemented a domain-specific programming language for stock and crypto modeling.
- Developed grammar definitions using textX and created a Python interpreter to execute scripts effectively.
- Created user-friendly stock and crypto declarations with operations like price changes, merges, and splits.
- Developed display features to present asset data, including prices, market caps, and supply metrics.
- Added a simulation mode to generate good, bad, or random days for specific stocks or the entire market.

Restaurant Flow Control

November 2024

- Designed a restaurant simulation to manage seating and customer flow with a producer-consumer model.
- Utilized a buffer to coordinate threads for seat request types, ensuring efficient resource allocation.
- Used mutexes and semaphores to ensure thread safety and prevent deadlocks in multithreaded operations.
- Critically analyzed resource utilization metrics to identify bottlenecks and improve overall system efficiency.
- Implemented a configurable CLI for testing various scenarios and analyzing the overall system performance.

Spam Detection AI Model

September 2023 – December 2023

- Collaborated with a team of five individuals to develop an AI Model designed to detect spam messages.
- Implemented a training set comprised of over 10,000 messages to effectively train our model using Python.
- Preprocessed the dataset thoroughly and implemented an algorithm based on Naive Bayes Classification.
- Achieved a remarkable 98% accuracy in detecting spam messages through rigorous debugging and testing.
- Created an engaging slide deck and confidently presented the final project to the Artificial Intelligence Club.

Additional Information

Honors: Dean's List (3/4 Semesters), High School Valedictorian & Faculty Selected Master of Ceremonies (June '22), 1st Place DECA International Career Development (Business Law & Ethics Roleplay)(May '21)

Organizations: Artificial Intelligence Club, ELeet Code Club, Undergraduate Student Research, Pickleball Club

Interests: Entrepreneurship, Lifting Weights, Pickleball, Basketball, Cooking, Chess, Camping, Running